ICAP Coding Manual

This document contains some background as well as the coding manual, involving the decision tree, coding rules, and examples along with fail examples that reflect borderline cases.

Background

This coding manual is based on the ICAP framework (Chi & Wylie, 2014) which can help distinguishing different levels of cognitive engagement of learners using overt behaviours. In their original conceptualisation, Chi and Wylie (2014) note that these behaviours refer to actions such as note taking, paraphrasing or self-explaining, each of which associated with varying levels of learning. Behaviours that involve higher levels of cognitive engagement are thus thought to lead to more learning. They distinguish the following four modes:

> ***Passive***: Learners receive information related to learning, e.g., through listening to a lecture or watching a video.
>
> ***Active***: Learners actively engage with instructional materials, e.g., by taking notes or highlighting sentences, activating prior knowledge.
>
> ***Constructive***: Learners produce learning output beyond what was originally provided in the learning materials (e.g., self-explanations, comparisons), inferring new knowledge by connecting and contrasting.
>
> ***Interactive***: Learners discuss issues with peers, creating new knowledge mutually. This requires that utterances be constructive and that sufficient turn taking occurs.

In discussion forums, a similar principle can be applied. Although not the same range of activities like in presence contexts can be identified, forum messages can still be examined regarding different levels of cognitive engagement based on a qualitative assessment of their content. This idea of using ICAP to study forum discussions was originally taken up by Wang et al. (2016) who adapted and applied the framework to study forum messages within Massive Open Online Courses (MOOCs). They introduced several adjustments to make the framework more suitable for forum discussions, including omitting the ***Passive*** category, since any forum comment already constitutes at least active behaviour, and adopting a more relaxed notion of what counts as ***Interactive*** behaviour. In addition, they differentiated between on- and off-task posts, only considering the cognitive engagement levels of posts if they are on-task. The final categories are *Off-task, Active*, *Constructive* and *Interactive*, with *Active* further

differentiated into *Active (Attention/Engagement)* and *Active (Paraphrase/Repeat)* and *Constructive* into *Constructive (Idea)* and *Constructive (Elaboration/Reasoning)*.

Our coding scheme is based on this conceptualisation by Wang et al. (2016). However, in the formal context they studied, learning activities and the ensuing discussion behaviours often evolve around pre-defined goals with clear task boundaries. In informal settings such as those that this coding scheme is targeted at (Citizen Science (CS) forums and Reddit), clear task definitions are not always present. At the same time, different project or platform contexts might influence what constitutes as "on-task". Accordingly, specific care must be taken when defining these task boundaries in our scenario. This will be more thoroughly discussed in the next section.

Setting

This coding scheme will be used to code discussion forum posts and comments from two informal settings: The Zooniverse CS platform and r/ask [...] subreddits on Reddit. On Zooniverse, users can participate e.g., by classifying chimpanzee images of wildlife cameras in the African rainforest or identify galaxies in telescope images, subsequently discussing their findings with others in an associated discussion forum. On r/ask [...] subreddits, users can ask questions and seek answers from other users. You will be coding the discussion posts from these two settings.

In this coding manual, we use posts and comments interchangeably, because they refer to the same unit within a thread.

Accordingly, what constitutes as on- vs. off-task can differ slightly per platform and project/subreddit. We will provide a short description of the project or subreddit in each case, such as: "in the following you will see comments from the discussion forum of the online citizen science project 'Chimp & See, where users classify chimpanzees in wildlife camera pictures from the African rainforest.". In our coding scheme, the categories are denoted as *on-topic* vs. *off-topic*. The distinction between on- and off-topic will be explained later.

Instructions and Categories

Your task is to code the cognitive engagement displayed in discussion forum posts of the given Zooniverse project or subreddit. For each post, you will additionally be provided with its immediate thread-level context: the thread starter (i.e., the initial post within the thread) and the previous post. On Zooniverse, the previous post is the one that occurred chronologically right before the target post. On Reddit, the previous post is the direct parent post (with a maximum depth of 2). In both cases, if the target post is the thread starter (or a top-level post on Reddit), no previous post is provided. The context should only provide additional information in case you find it hard to accurately code the target post on its own. In this case, first consider the thread startern, and then the previous

comment. There is, however, no guarantee that this will clarify the target comment's context.

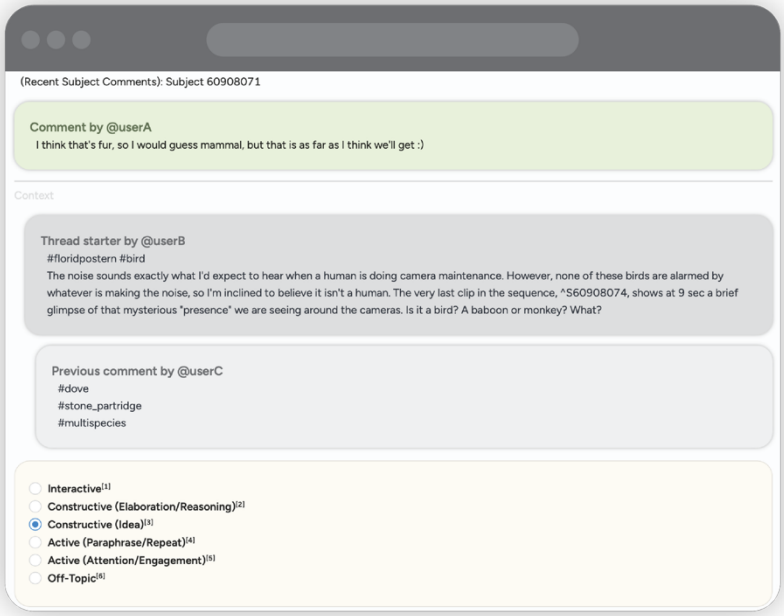


Fig. 1. The coding environment you will see. In addition to the target comment (green), you will be provided with the thread starter and previous comment.

The coding is limited to the textual content of each post. Although some posts might contain implicit or explicit references to visual elements, the images themselves are not available for this task. You should only focus on the textual content itself.

The unit for coding is the whole post. Typically, posts do not become longer than 80 words.

For each post and context, we also provide the username of who wrote it. This can help to determine whether there are explicit references between users, which is particularly relevant for the *Interactive* category (more on that later).

Decision Tree

Before going into detail about the specific categories, we will provide the decision tree that you should follow while deciding on each category. The first decision you should take while coding is whether the target post is *on-* vs. *off-topic*. If it is *on-topic*, proceed with distinguishing its level of cognitive engagement. If it is *off-topic*, label it as *off-topic* and

proceed to the next post. In the next section, we will explain each category in more detail and give examples.

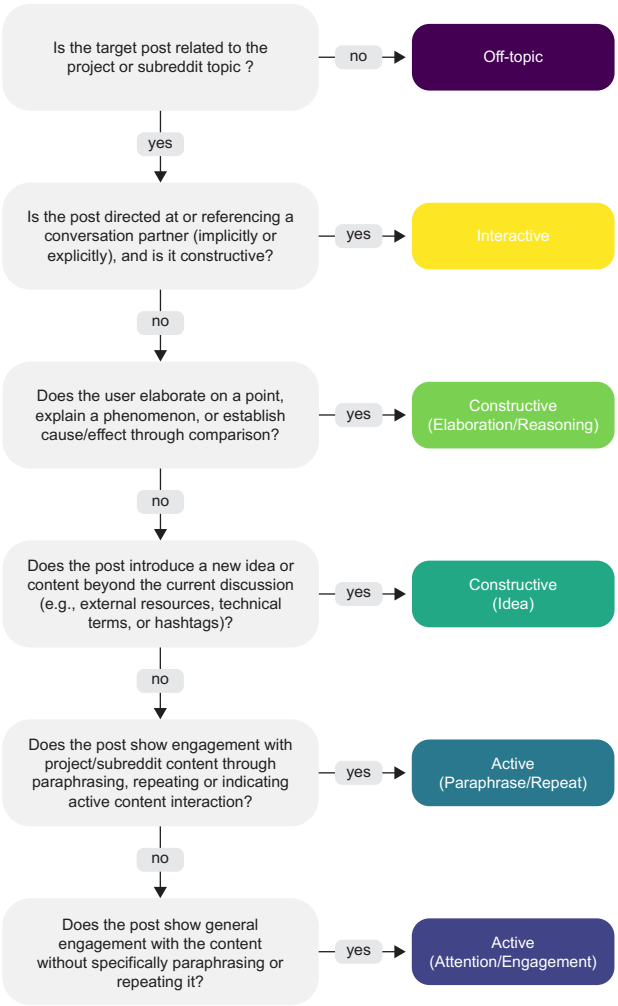


Fig. 2. Decision tree for the coding manual roughly based on Wang et al. (2016)

On-topic

If the user talks about project or subreddit-related content, it is considered to be on-topic. This can be related to the content of classifications, related questions, or general subreddit-related questions. The decision whether something constitutes as project- or subreddit-relevant should be made according to the short project or subreddit description you will be provided with. There will be no frequent switches between projects, so you can get familiar with the specific setting or project/subreddit until you finish your assigned batch of posts.

If the post indicates that the user engaged with project- or subreddit-relevant content (i.e., on a meta-level, not talking about the content itself), it should also be coded as on-

topic. An example for this would be if the user talks about how he classified images for the whole evening.

Note that social talk, humour, or meta-comments about participation also qualify as on-topic when they demonstrably reference or engage with project or subreddit content. For instance, a joke that revolves around a specific classification or a compliment that explicitly acknowledges a content-related explanation would be considered on-topic, as the content relation is evident.

If the content-relation cannot be inferred from the target comment itself, but its context is project or subreddit-related, it also makes the target comment on-topic.

Example 1 (Zooniverse: Exoplanet Explorers)

Target comment (@userA): *"I'll come back in a few days time. Seems pointless to classify at present."*
Thread starter (@userB): *"Im having a real problem. Everything I say yes to, even VARIABLE stars or blank data's says 'Simulated'. Anyone else having this problem?"*
Previous comment (@userC): *"Hi @userD. – Thank you."*

In this example, the target comment is on-topic because, although not talking about project content itself, the user still indicates that they dealt with the content. Also, the context is not really needed for this decision.

Example 2 (Reddit: r/askastronomy)

Target comment (@userA): *"Its 99.9% nothing and .1% "stuff" inbetween the planets. Fun fact for you: How close are rocks are from one another in the asteroid belt you think? The answer is 1,000,000 km (600k miles), roughly double the distance of earth/moon."*
Thread starter (@userB): *"Why don't people don't talk much about Ceres? Like I don't understand. There is literally an entire dwarf planet fully round in shape, close to us. Pluto takes all the fame instead."*
Previous comment (@userC): "*I've been wondering the same thing. There's so much extra stuff in the solar system that people don't seem to care about. I remember seeing Ceres as well as the Jovian moons and realizing how much more complicated and interesting the solar system is than just the planets themselves."*

In this example, the target comment is on-topic as it's related to the subreddit's topic. Although it is not a direct answer to the original question posed in the thread starter, it can still be labelled as on-topic.

Fail example (Zooniverse: Galaxy Zoo)

Target comment (@userA): *"Username checks out, what an awesome galaxy pun! Wish I came up with it..."*

Thread starter (@userB): *"Hello fellow Zooites. Great to see so many new faces. Who wants to introduce themselves first?"*
Previous comment (@userC): *"Hello from Arkansas, glad to be participating. Best, @userC."*

In this example, the target comment talks about user names with a distant relation to the project's topic. However, no actual content-relation can be identified. On top of that, the comment is in a thread that is about users introducing themselves. Accordingly, it should be labelled as off-topic.

Off-topic

If the target post does not contain content-relevant references, it should be labelled off-topic. This includes self-introductions, technical or platform-related issues (e.g., display problems, loading errors), moderation posts, platform complaints, humour or jokes without content engagement, and social talk without content-relation. The following examples illustrate this more:

Example 1 (Zooniverse: Galaxy Zoo)

Target comment (@userA): *"My laptop can't display the website properly"*
Thread starter (@userB): *"I can't seem to load Subject 67890. Is anyone else having trouble?"*
Previous comment (@userC): *"It's loading fine for me. Maybe try clearing your cache?"*

In this example, the target comment is related to a technical issue, which lends itself to the "off-topic" label. While comments like this might imply in certain cases that the user actively tried to participate in the project to some extent, the content of the comment itself does not show any content-relation, and considering that the user talks about technical issues, we label it as "off-topic".

Example 2 (Zooniverse: Exoplanet Explorers)

Target comment (@userA): *"Just discovered this. Totally addictive. What fun. Hugs from Sydney"*
Thread starter (@userB): *"Hi all. I'm an Astronomer from the UK. Just found out about this project yesterday. After 20 years in related research I figured now is my time to shine and help the scientists!"*
Previous comment (@userC): *"they are gonna be looking at a black hole in the next few days for the first time, though we wont get a picture for about a year. So many cool thinghs happening almost every month. just loving this."*

In this example, we cannot tell directly from the target comment whether it's content related. Looking more closely at the context, it seems to be a thread about self-introductions, which would indicate the comment is off-topic. Further, considering the previous comment, it seems that the comment implicitly references what @userC said

earlier about general research on black holes. While this is astronomy-related, it is not about the content of the project itself. Thus, the target comment should be labelled "off-topic".

Fail example (Reddit: r/askastronomy)

Target comment (@userA): *"nice answer! this way, I understand it!"*
Thread starter (@userB): *"Why do some stars appear to twinkle more than others?"*
Previous comment (@userC): *"It's mainly about how much atmosphere the light has to pass through. Stars near the horizon twinkle more because their light travels through a thicker layer of air, which causes more refraction. Stars overhead have a shorter path, so they look steadier."*

In this example, the target comment answers to a previous comment using a form of social talk. Considering only the target comment, we could come to the conclusion that it is off-topic because no actual content relation is made explicitly. However, looking at the previous comment, we see that the user actually engaged with the content by uttering that they understood it given the explanation from the other user. Accordingly, as this social talk is content-related, we label it as "on-topic".

Interactive

A post should be labelled as *Interactive* if fulfills two criteria. Firstly, it needs to be constructive (i..e., elaborating on something, reasoning about). Secondly, it needs to be directed at a conversational partner, either implicitly or explicitly, which typically involves building upon or elaborating on the ideas of the partner. As such, it fulfills the ICAP definition of the category, where new knowledge is created through co-inferring. As mentioned, the post does not need to contain explicit references to others as implicit ones suffice. This will become more clear in the examples.

Example 1 (Zooniverse: Galaxy Zoo)

Target comment (@userA): *"I see your point about the width, @userC, but it's perfectly straight and has a uniform brightness, which is uncharacteristic of any galactic structure. I think it has to be an artifact."*
Thread starter (@userB): *"I think this is a satellite trail, not a feature of the galaxy."*
Previous comment (@userC): *"I'm not so sure. It seems too wide for a typical satellite."*

In this example, the target comment directly addresses @userC, acknowledges their point, and then presents a counter-argument with specific reasoning. This fulfills the first criterion for it to be at least constructive, and the second one for being directed at a conversational partner and building on their ideas.

Example 2 (Reddit: r/askastronomy)

Target comment (@userA): *"That's a good point about the foreground star. Considering this, what if we pull up the SDSS catalog entry for that region? If there's a known star at those coordinates, that would settle it pretty quickly."*
Thread starter (@userB): *"I was looking at NGC 3227 in my telescope last night and the core looks weirdly elongated. Could this be a sign of a merger event?"*
Previous comment (@userC): "It does look disturbed, but that elongation could also be a foreground star overlapping with the galaxy's core. That can be tricky to distinguish visually.*"*

In this example, the target comment directly addresses @userC, acknowledges their point, and then presents a counter-argument with specific reasoning. This fulfills the first criterion for it to be at least constructive, and the second one for being directed at a conversational partner and building on their ideas.

Fail example (Zooniverse: Exoplanet Explorers)

Target comment (@userA): *"Interesting. I think we should also check if there's any stellar variability that could mimic a transit signal. That's always worth ruling out."*
Thread starter (@userB): *"I found a really clean-looking dip in the light curve for KIC 8462852. The depth looks consistent with a Jupiter-sized planet. Anyone else see this?"*
Previous comment (@userC): "I noticed the dip too, but the spacing between transits seems irregular. That could point to a multi-planet system rather than a single companion.*"*

In this example, the target comment first appears to be Interactive because it references the other user's point. However, it does not directly engage with it or elaborate on it further, instead, it proposes another, new idea. Accordingly, it rather fits the Constructive (Idea) category.

Constructive (Elaboration/Reasoning)

If the target comment does not show *Interactive* behaviour, you should decide whether it is *Constructive*. The first category of relevance is *Constructive (Elaboration/Reasoning)*. If the user provides an argumentation for a point (e.g., through elaboration, constructing a cause- and effect relationship by comparing two aspects), it should be coded with that label. An important aspect in this category is the existence of actual argumentative and reasoning traces, opposed to much shorter utterances. This will become clear in the following examples:

Example 1 (Zooniverse: Exoplanet Explorers)

Target comment (@userA): *"This is a #goodtransit. The #transiting is visible in all data segments, the dip is U-shaped with a flat bottom, and the star's metrics confirm it's not an eclipsing binary."*

Thread starter (@userB): *"Found a potential transit. #transit"*
Previous comment (@userC): "*Looks interesting. Can you be more specific?"*

In this example, the user in the target comment elaborates on the original "transit" claim by providing multiple pieces of specific evidence and gives reasons why this evidence supports the conclusion. As such, they elaborate on a point, qualifiying this comment as Constructive (Elaboration/Reasoning).

Example 2 (Reddit: r/askastronomy)

Target comment (@userA): *"That dimming phase explanation makes sense when you look at it more closely. Betelgeuse is a semi-regular pulsating variable, so its brightness fluctuates as the star expands and contracts. The current dimming is consistent with the overlap of its roughly 400-day and 5.9-year pulsation cycles, which together can produce these more noticeable dips"*
Thread starter (@userB): *"I noticed that Betelgeuse looks noticeably dimmer than usual tonight. What could be going on?"*
Previous comment (@userC): "*I think it's just going through one of its dimming phases."*

The target comment elaborates on the idea from the previous comment by connecting multiple points to construct a logical explanation that supports the initial suggestion. As it deepens an existing point through reasoning, it is considered Constructive (Elaboration/Reasoning).

Fail example (Zooniverse: Galaxy Zoo)

Target comment (@userA): *"Those definitely look like tidal features to me. For me the unusual spiral arms are telling."*
Thread starter (@userB): "*Subject 58812 has some unusual spiral arms. What's going on with this one?"*
Previous comment (@userC): "*I think those arms are tidal tails from a recent interaction with a nearby companion galaxy."*

While the comment might appear elaborative at first, it merely repeats what was already said in the thread. As such, it does not show any reasoning or elaboration, nor does it introduce something new. Accordingly, it should not be labelled Constructive (Elaboration/Reasoning) but rather Active (Paraphrase/Repeat).

Constructive (Idea)

If the comment does not contain any reasoning or elaboration but introduces aspects or topics that go beyond the current discussion, it should be labelled as *Constructive (Idea)*. Judging from the target post and its surrounding context alone, it can be tricky to identify what constitutes as the introduction of new aspects. At the same time, the topic of a subreddit or Zooniverse project is also not always clearly delineated. Accordingly, we

restrict this category to apply to aspects the user introduces that go beyond the current discussion, as we can judge it given the target comment and its surrounding context.

In most cases, this refers to links, hashtags or other external resources that the user introduces into the discussion. Similarly, if the target post is the thread starter and the user introduces a new hashtag for a classification or puts forward an idea regarding a specific subject, it would also qualify as *Constructive (Idea)*. Generally, this category refers to posts that contain *Constructive* behaviour without elaborating on it.

Example 1 (Zooniverse: Exoplanet Explorers)

Target comment (@userA): *"The following link explains why V-shaped dips often indicate an eclipsing binary system rather than a transiting planet. [link]"*
Thread starter (@userB): *"This is a very deep transit for such a small star. Could it be a very large planet?"*
Previous comment (@userC): "*It's possible, but the V-shape of the dip is suspicious.*"

The user in the target comment introduces external information to support a new interpretation of the data, which adds new information to the discussion, although it itself does not contain elaborations or reasoning traces. As such, it should be labelled Constructive (Idea)

Example 2 (Zooniverse: Galaxy Zoo)

Target comment (@userA): *"Looks like an interaction @ 7:00 that is extending to the left of the photo … very interesting … Subject 23456"*
Thread starter (@userB): *Target comment is the thread starter*
Previous comment (@userC): *N/A*

The target comment is the thread starter, initiating a new thread by pointing out a specific, novel feature in a specific location of an image, thereby introducing a new idea for discussion. Accordingly, it should be labelled Constructive (Idea)

Fail example (Reddit: r/askastronomy)

Target comment (@userA): *"Exactly. Jupiter's apparent magnitude is around -2.7 compared to Saturn's +0.5, and that gap makes sense when you consider that Jupiter's disk has roughly 3.5 times Saturn's angular diameter at opposition. The larger apparent size means way more reflected light reaching us."*
Thread starter (@userB): "*Why does Jupiter appear so much brighter than Saturn even though Saturn's rings should reflect a lot of light?"*
Previous comment (@userC): *"It's mainly because Jupiter is much closer to us and also much larger, so it reflects a lot more sunlight overall."*

At first, the target comment seems as if it introduces new aspects (data) that were not present before. However, the user uses these data to construct their argument, thereby

elaborating on a point and showing reasoning. As such, it should not be labelled as Constructive (Idea) but rather as Constructive (Elaboration/Reasoning)

Active (Paraphrase/Repeat)

If the comment does not show *Constructive* behaviour, it might be *Active*. The first category you should consider is *Active (Paraphrase/Repeat)*. It applies if the user shows active engagement with the project or subreddit content through repeating or paraphrasing existing information, yet does not introduce new aspects or elaborate on issues. The information can be project/subreddit related, or relate to the discussion thread itself.

This category also applies when the user asks questions regarding content-related aspects.

Example 1 (Zooniverse: Galaxy Zoo)

Target comment (@userA): *"This galaxy appears to have so many very faintly visible arms. Its stunning!"*
Thread starter (@userB): "*Look at the faint structures in this one. What a beauty. Subject 55678"*
Previous comment (@userC): *"I agree, it looks like there are a lot of arms."*

In this example, the user who wrote the target comment paraphrases the previous user's observation using similar descriptive language, affirming the perception without adding new information. The label is thus Active (Paraphrase/Repeat).

Example 2 (Zooniverse: Exoplanet Explorers)

Target comment (@userA): *"#possible"*
Thread starter (@userB): "*Found an interesting signal here. Looks like a possible transit."*
Previous comment (@userC): *Previous comment is the thread starter*

In this example, the user uses a common project hashtag to briefly agree with the initial finding described in the thread starter. As such, they essentially repeat the claim "possible transit", thus not adding new knowledge nor elaborating on a point. The corresponding label is thus Active (Paraphrase/Repeat).

Fail example (Reddit: r/askastronomy)

Target comment (@userA): *"pretty rough framing on this one."*
Thread starter (@userB): "*Here's my first attempt at capturing the Orion Nebula with a 6-inch reflector. Any tips for improving the detail?."*
Previous comment (@userC): *"Not bad for a first try! The core is nicely exposed, but the outer nebulosity is getting lost. You could try stacking more sub-exposures to bring that out."*

In this example, you might be tempted to label the comment as Active (Paraphrase/ Repeat) because it seems to reflect an observation about the image. However, the user does not paraphrase or repeat any specific content from the thread or the image itself, only offering a brief evaluative reaction without describing, restating or indicating active interaction with any particular aspect of what they see. This rules out the category. However, it also clearly shows that the user looked at the image and thus engaged with the content to some extent. Accordingly, Active (Attention/Engagement) fits more.

Active (Attention/Engagement)

If the target post does not show paraphrasing or elaboration of project/subreddit content but does indicate that the user engaged with the content, it should be labelled as *Active (Attention/Engagement)*. This particularly refers to situations where the user explicitly or implicitly describes how they engaged with the project/subreddit content, yet not go on the content-level (e.g., "I classified for the whole evening today. Felt great"). Although such posts do not contain substantial content, they still show that the user engaged with the project/subreddit content.

Example 1 (Zooniverse: Exoplanet Explorers)

Target comment (@userA): *"Starting a new search for planets right now! Already looked through all the documentation, very excited to finally start."*
Thread starter (@userB): *"Welcome to all new Exoplanet Explorers! What are you up to today?"*
Previous comment (@userC): "*Hi everyone, I just completed the tutorial!"*

Here, the user describes that they looked through the documentation of the project, indicating that they engaged with the content, at the same time saying that they are about to start classifying. While nothing was said on the content level, it still shows engagement with the project, thus Active (Atttention/Engagement)

Example 2 (Zooniverse: Galaxy Zoo)

Target comment (@userA): *"Just saw the new images."*
Thread starter (@userB): *"The new data release is live!"*
Previous comment (@userC): *Previous comment is the thread starter*

The user describes that they saw the newly released images, showing engagement with the project content. However, they do not descibe them, paraphrase something that was said nor do they elaborate on them. Accordingly, Active (Attention/Engagement) fits.

Fail example (Reddit: r/askastronomy)

Target comment (@userA): *"I was out all night too but honestly I just sat there with my dog enjoying the quiet. Best way to spend a summer evening."*
Thread starter (@userB): *"Did anyone else catch the Perseid meteor shower last night? I counted about 40 per hour from my location."*
Previous comment (@userC): "I had a great view from the mountains, counted even more around 2 AM when the radiant was highest."

Now this is a tricky one. The thread is content-related. The target post itself is describing engagement too, however, this engagement is not directed towards the project content (in this case the topic of the actual discussion) but rather on something that is off-topic. Accordingly, it does not qualify as Active (Attention/Engagement) but rather as off-topic.

References

Chi, M. T. H., & Wylie, R. (2014). The ICAP framework: linking cognitive engagement to active learning outcomes. *Educational Psychologist*, *49*(4), 219–243. https://doi.org/10.1080/00461520.2014.965823

Wang, X., Wen, M., & Rosé, C. P. (2016). Towards triggering higher-order thinking behaviors in MOOCs. *Proceedings of the Sixth International Conference on Learning Analytics & Knowledge*, 398–407. https://doi.org/10.1145/2883851.2883964